SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	Dsaise
Project / Product Name	HalaTuju
Final Live Deployment URL	https://halatuju.replit.app
Codebase Repository URL	https://github.com/shenenleow10521/SdcHackathon
Name (Leader)	Leow Shen En
Email & phone (Leader)	leowshenen@gmail.com, 018-2335269

2. Project Description

[Maximum 250 words]

HalaTuju shows a worker which of their skills AI is taking over, what to learn next, and whether their plan still holds as AI changes. A person uploads a CV, pastes text or answers a short task quiz. A privacy scrubber removes personal details before any AI model sees the text. The app then matches their experience against all 13,939 ESCO skills.

Every skill gets an AI-exposure score from our own model, trained on human task ratings on a RunPod GPU. The worker sees three groups: skills at risk, skills AI assists, and skills that stay theirs. They can choose to stay and become AI-ready in their current job, or move. For a move, HalaTuju ranks roles from 3,039 ESCO occupations, explains each match score and its skills gap, and builds an ordered plan fitted to their weekly hours, using free or low-cost courses that were checked by hand.

Every plan is replayed in 1,000 simulated futures and shown under Today and 2030 Slow, Steady and Fast AI adoption, labelled as estimates. Every job and skill carries an SDG tag with a reason. The plan view shows how the chosen path supports each goal.

A researcher agent reads 74 trusted sites. Three AI agents vote on each new claim, and a person decides split votes. Users affected by an approved change choose to keep or switch their route. Fixed formulas, not the language model, set every score. HR teams see an opt-in team skills view.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Browser app (JavaScript, HTML/CSS, English and Bahasa Melayu): onboarding (CV, pasted text or task quiz), My skills, Explore (career map where distance is time to job-ready), Plan (route, futures chart, Today vs 2030 Slow/Steady/Fast), My goal (weekly check-ins, proof of each skill), and For HR (opt-in team progress).

API: Node 22, TypeScript, Express 5, served from one Replit Reserved VM. Main modules:

• Profile intake: local Tesseract OCR, privacy scrubber, then an LLM splits the CV into phrases. bge-m3 embeddings on RunPod Serverless find candidate ESCO skills, and a second LLM call confirms skills and the current job. The user confirms or removes each skill.

• Scoring engine (shared/engine.js, shared by server and browser): fixed formulas for match score (skill overlap weighted by level and exposure), route time and RM cost, scenario shift, and the 1,000-future Monte Carlo stress test.

• ESCO services: 3,039 occupations, 13,939 skills, 126,051 links, O*NET tasks with human exposure ratings, ILO WP140, DOSM wages and MyCOL flags. Routes can be built for any ESCO job, with cross-field bridges reviewed before use.

• Trend pipeline (Road Watch): a scheduled researcher searches 74 trusted domains (Tavily, RSS, arXiv, OpenAlex, GDELT). Quotes are checked word for word. Three voter agents on models from two vendors (evidence checker, skill matcher, sceptic) vote; 2–3 of 3 apply, 1 of 3 goes to a person, 0 is rejected. Every vote is logged. Approved changes create a new knowledge version (which can be undone), and affected users get a keep-or-switch notice.

• SDG mapping (shared/sdg.js): hand-written reasons for catalogue items, a rule-based mapping for every ESCO job and skill, summed into the user's path contribution.

AI models: KrackedDevs proxy (main gpt-5.5, reviewer DeepSeek-V4-Pro, voters gpt-5.5 / gpt-6-sol / DeepSeek-V4-Pro), with Google Gemini (gemini-3.5-flash-lite / gemini-3.1-flash-lite) as fallback; BAAI/bge-m3 on RunPod Serverless; our ridge exposure model trained on RunPod. LLMs only read text and explain. They return database IDs, never free-text jobs or courses.

Data: Replit PostgreSQL 16 (profiles, versioned knowledge, signals, votes, goals, audit log). Sign-in: Replit-managed Clerk, with guest access. CI: GitHub Actions (type-check and 543 automated tests).

Diagram: Appendix A.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

What we changed and why. The Phase 1 prototype answered "where can I move?" but users first ask "will AI take my job, and which of my skills?" We changed the order of the product to stay first, then move.

Built and deployed in the final round (30–31 Oct):

1. My skills outlook: every confirmed skill sorted into at risk, AI-assisted and still yours, with one action per skill.

2. Stay AI-ready track and My goal: weekly check-ins, proof of each skill, a skills record, and opt-in team progress for HR.

3. Routes for any ESCO job: Phase 1 had routes for 16 sample jobs. Now any of 3,039 occupations works, with AI-era skills, cross-field bridges (reviewed before use), practice tasks and 45 hand-checked learning resources (38 free) (every URL opened on 30 Oct).

4. Map by time: distance now means months to job-ready (3/6/12/24-month rings), and size means match.

5. Three-agent vote: one AI reviewer replaced by three voters on two vendors' models, with an audit log; split votes go to a person.

6. Scenarios: Today vs 2030 Slow/Steady/Fast side by side, driven by an adoption-speed parameter, with what shifts spelled out; a futures chart.

7. SDG mapping for every ESCO job and skill, with a reason, and the path contribution shown in Plan.

8. No-CV task quiz for thin profiles (4,017 O*NET tasks).

Problems we solved along the way: Gemini free-tier quotas led us to the KrackedDevs proxy, which also gives the vote panel models from two vendors. Bridge suggestions between unrelated fields were sometimes wrong, so we added validation scripts and human review. A single reviewer could be fooled by vendor marketing, so we added a sceptic voter.

Tests grew from 418 to 543 (36 files, all passing on 31 Oct).

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

The data and evidence you provide in this section will be used to assess SDG Impact, Technical Execution and Innovation scores. Vague claims or missing verification will weaken your score across all three categories.

Metric Category	Your Response
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	SDG 8.5 and 4.4 (8.2 secondary). Against the Phase 1 KPIs: **(1) Current-job identification:** target 95% on a fresh blind set; result **97.2% (35/36)** on held-out synthetic CVs (95% CI 85.8–99.5). **(2) Personal details removed before AI calls:** target 99%; **99.4% (154/155)** after fixes, not blind (blind first run 82.6%). A fresh blind set was not run in the final round. **(3) Users finish a costed route within 60 s:** target 80%; **not measured with users** in the hackathon window. On the live app, the sample persona's costed route (3 steps, RM 0, ready in 14 weeks at 6 h a week) is returned by one API call. **(4) Pilot participants with an advisor-approved plan:** target 70%; **not run**, because a consent-based pilot with real users was outside the hackathon window. **New in Phase 2:** coverage grew from 16 to 3,039 occupations with routes; 38 of 45 hand-checked learning resources are free; none is marked HRD Corp-claimable, because no course could be confirmed on an official HRD Corp page.
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	CV benchmark: scripts/eval-cv.ts over 50 synthetic CVs. Answer keys written by an independent annotator, and their sha256 is stored in each result file (eval/results/cv-full-*.json, eval/RESULTS.md). Skill precision was judged blind with decoys (scripts/esco-judge.ts). Vote replay: scripts/eval-votes.ts (eval/VOTES.md; only 1 human-decided signal so far, so it proves nothing about accuracy).

Metric Category	Your Response
	Plan, cost and scenario figures come from the live API (/api/plan, 31 Oct 2026); the 1,000-future result from stressTest in shared/engine.js run on live data. Every learning URL was opened and checked on 30 Oct (db/seed/learning.json). 543 automated tests in CI, including a fairness test.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	Context: 1,360,100 Malaysian clerical wage recipients in 2025 (OpenDOSM), a group with high GenAI exposure (ILO WP140). First channels: university career centres and TVET providers (advisor approval flow), then HR teams (opt-in team view, keep-or-switch). A worker gets a sourced, costed plan in minutes instead of a one-off assessment that goes stale. SDG tags change choices: the ranking can favour roles that support the user's chosen goals.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	Free for individuals. Paid per-seat HR and career-centre dashboard. The engine can be called as an API by learning-management systems. Pricing (pilot assumptions from our business model): RM 300 per worker a year; fixed cost RM 24,452 a month; RM 24.50 contribution per seat a month, so break-even at about 1,000 seats. Variable cost stays low because scores and routes are fixed formulas with no model call, LLM answers are cached, and RunPod bge-m3 scales to zero (the full training run of our exposure model cost about US$0.04 on an RTX 4090). The sample plan's three learning resources cost RM 0. Trend data refreshes daily (10 skills per run, all skills every 3 days). Privacy under PDPA: scrubbed before AI calls, CV files not stored, user can delete their profile. Fairness: tests/fairness.test.ts runs on every push. The same CV with a different name, gender, age or religion gives the AI model identical text, and the app returns identical skills,

Metric Category	Your Response
	levels, job matches and routes, both with the AI model and on the basic fallback path.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

Match score (shared/engine.js). Every score on screen is this formula, not a model opinion. Coverage is the level you have over the level needed (max 1); 0.4 for a skill you can bridge to from one you own; 0.3 for a skill AI has taken over in the chosen scenario.

```
function match(job, extraReq = []) {
  const r = [...(graph.req[job] || []), ...extraReq];
  if (!r.length) return 0;
  const num = r.reduce((s, q) => s + weightOf(q.skill, q.weight) * coverage(q.skill, q.need), 0);
  const den = r.reduce((s, q) => s + weightOf(q.skill, q.weight), 0);
  return Math.round(100 * num / den);
}

// Scenario: one AI-speed parameter s in [-1, 1] (Slow, Steady, Fast) moves every value on its 2026 -> 2030 path
export function aiInterp(arr, f, s) {
  if (s <= 0) return arr[0] + (arr[1] - arr[0]) * f * (1 + (1 - PARAMS.slowFactor) * s);
  return arr[0] + ((arr[1] - arr[0]) * (1 - s) + (arr[2] - arr[0]) * s) * f;
}
```

Output from the final build (31 Oct 2026, sample persona: accounts-payable clerk, 6 h a week, deadline 15 Feb 2027; target: ESG & Carbon Reporting Analyst):

```
Today (2026):   match 57%, 14 weeks, RM 0; AP -> reconciliation -> GHG accounting
2030 Fast AI:   match 51%, 13 weeks; AI takes over reconciliation, so the route
                switches to AP -> audit trail -> sustainability disclosure
2030 Slow AI:   match 57%, 14 weeks; same route as today
1,000 futures:  job-ready by the deadline in 719 of 1,000; the misses are the
                futures where study time drops below 5.5 h a week
```

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Who first: three Malaysian university career centres and two TVET providers in a consent-based pilot, then HR teams in shared-services centres (Penang, Klang Valley), where clerical and finance roles are most exposed.

Scaling the data: the trend pipeline already runs on a schedule over 74 trusted domains. With funding we would add licensed Malaysian job-ad feeds and Bahasa Melayu news, and cover all ESCO skills each cycle instead of 10 per run. eMASCO/MASCO codes would sit beside ESCO so results match Malaysian statistics.

Scaling compute: bge-m3 on RunPod Serverless scales from zero to many workers. Scoring runs in fixed formulas with no LLM call, so the cost per extra user is mostly storage. LLM calls happen once per CV and once per evidence check, not on every screen.

Trust at scale: every model decision is logged. Human review stays on split votes and route-changing evidence. We would add a published fairness test each release and a PDPA-compliant data-retention policy reviewed by a partner's legal team.

Partners: HRD Corp for claimable programmes; TalentCorp for MyCOL updates; Rotary and XMUM research links for pilot sites and an outcome study (did users reach an approved plan by their deadline?).

Business model: free for individuals; per-seat licence for HR and career-centre dashboards; an API for learning-management systems.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.

☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.

☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.

- [ ] The final pitch deck has been exported and saved as a flattened PDF file.
- [x] All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Node.js, Express, TypeScript, Tesseract.js, pg, PGlite, 3d-force-graph, three.js, GSAP, Clerk SDK	Code	MIT / Apache-2.0 / BSD / GreenSock standard
BAAI/bge-m3	Model	MIT
Google Gemini (gemini-3.5-flash-lite, gemini-3.1-flash-lite)	Model (API)	Gemini API terms
GPT-5.5, gpt-6-sol, DeepSeek-V4-Pro via KrackedDevs proxy	Model (API)	Provider and sponsor terms
ESCO v1.2.1, ESCO–O*NET crosswalk	Dataset	CC BY 4.0 / EC reuse policy
O*NET tasks, ratings, Detailed Work Activities	Dataset	CC BY 4.0
ILO WP140; Anthropic Economic Index; GPTs are GPTs labels	Dataset	CC BY 4.0 / CC BY / MIT

DOSM/OpenDOSM; MyCOL 2024/25	Dataset	CC BY 4.0 / facts cited
Learning resources (YouTube, Microsoft Learn, Open Library, IIA, course providers), links only	Dataset (links)	Linked only; provider terms

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



Date: 31 October 2026

Appendix A: System block diagram

